Raghavendra Kulkarni

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Education

Northeastern University — Boston, MA
Master of Science in Computer Science (GPA: 3.86) — *Aug. 2024 – Dec. 2026*

Visvesvaraya Technological University — Bengaluru, India
Bachelor's in Information Science and Engineering — *Aug. 2018 – July 2022*

Experience

Software Engineer Intern — Jan. 2026 – Aug. 2026
Vicor Corporation — *Andover, MA*

- Developed a production BOM Viewer and REST APIs using **React**, **Java**, **Spring Boot**, and **Oracle**, batching SAP requests and adding server-side caching for workflows processing **10,000+** daily transactions.
- Refactored a data-intensive PCB app using lazy loading and asynchronous JavaScript, cutting page latency from **80s to 2s** for production workflows used by **200+** engineers.
- Automated **Docker** deployments via **Jenkins CI/CD**, instrumenting services with Prometheus/Grafana and automated failure alerts to speed up production troubleshooting.
- Engineered **JUnit 5/Mockito** test suites achieving **95%+** coverage for critical backend services, reducing regression defects by **50%** and enabling safer releases.

AI Software Engineer (Research Assistant) — Sep. 2025 – Dec. 2025; Sep. 2026 – Present
DASH AI Hub, Northeastern University — *Boston, MA*

- Led development of an AI grading platform using **Next.js**, **Flask**, and **LangGraph**, reducing essay grading time from 8+ hours to under 30 minutes and driving adoption across 10+ faculty members.
- Developed a semantic RAG pipeline with **LlamaIndex**, **Qdrant**, and metadata filtering, validating response quality through controlled experiments and faculty feedback.
- Scaled AI inference from **3 to 40 workers** using shared Whisper services and **RabbitMQ**, eliminating GPU memory failures; automated AWS and on-premises deployments with **Docker** and **GitHub Actions**.

Software Engineer — Oct. 2022 – Aug. 2024
Cognizant Technology Solutions — *Bengaluru, India*

- Implemented **React** interfaces and **Python** REST services for SAP's Service Provider Cockpit, enabling consultants to manage server provisioning and lifecycle workflows.
- Extended the Cockpit's Python automation layer across **AWS, Azure, and SAP HANA Cloud**, eliminating **40+ hours** of manual setup per environment deployment.
- Migrated core OneCognizant modules from **.NET to Java and Angular**, optimizing SQL queries to cut page latency by **40%** and eliminate recurring timeout issues.

Software Engineer Intern — *Apr. 2022 – Sep. 2022*
Cognizant Technology Solutions — *Bengaluru, India*

- Built a service-booking platform using **Angular** and **Spring Boot**, integrating **JWT** authentication and **PostgreSQL** for secure booking workflows.

Projects

Learnbot | *React, Flask, Pinecone, Claude, RAG*

- Built a React/Flask RAG teaching assistant used by 50+ finance students, combining Canvas content ingestion, Pinecone semantic search, and Claude to deliver course-specific guidance within academic-integrity constraints.

DASH API | *React, Flask, REST APIs, Linux, GPU Infrastructure*

- Built a self-service platform for accessing open-source LLM models on an on-premises GPU server, enabling faculty and students to reserve time slots and receive time-bound access through REST APIs or SSH.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL
Frameworks: Spring Boot, React, Next.js, Angular, Flask, FastAPI, Node.js, JPA/Hibernate, JUnit, Mockito
AI/ML & Databases: LangGraph, LlamaIndex, Qdrant, Pinecone, Claude, PostgreSQL, MongoDB, Oracle, Redis
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, RabbitMQ, Jenkins, GitHub Actions, Linux, Git
Core Competencies: Data Structures & Algorithms, Microservices, Distributed Systems, Agile, Git, Version Control